Representation of the Visual Field in the Primary Visual Area of the Marmoset Monkey: Magnification Factors, Point-Image Size, and Proportionality to Retinal Ganglion Cell Density

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ABSTRACT

The primary visual area (V1) forms a systematic map of the visual field, in which adjacent cell clusters represent adjacent points of visual space. A precise quantification of this map is key to understanding the anatomical relationships between neurons located in different stations of the visual pathway, as well as the neural bases of visual performance in different regions of the visual field. We used computational methods to quantify the visual topography of V1 in the marmoset (*Callithrix jacchus*), a small diurnal monkey. The receptive fields of neurons throughout V1 were mapped in two anesthetized animals using electrophysiological recordings. Following histological reconstruction, precise 3D reconstructions of the V1 surface and recording sites were generated. We found that the areal magnification factor (M_A) decreases with eccentricity following a function that has the same slope as that observed in larger diurnal primates, including macaque, squirrel, and capuchin monkeys, and humans. However, there was no systematic relationship between M_A and polar angle. Despite individual variation in the shape of V1, the relationship between M_A and eccentricity was preserved across cases. Comparison between V1 and the retinal ganglion cell density demonstrated preferential magnification of central space in the cortex. The size of the cortical compartment activated by a punctiform stimulus decreased from the foveal representation towards the peripheral representation. Nonetheless, the relationship between the receptive field sizes of V1 cells and the density of ganglion cells suggested that each V1 cell receives information from a similar number of retinal neurons, throughout the visual field. J. Comp. Neurol. 521:1001–1019, 2013.

INDEXING TERMS: *Callithrix jacchus*; visual cortex; receptive fields; computational maps; primate; peripheral scaling

Each half of the visual field is represented in the contralateral primary visual cortex (V1) in a continuous and systematic way, which defines a visuotopic map (Talbot and Marshall, 1941; Daniel and Whitteridge, 1961). The geometry of this map dictates the number of neurons that are dedicated to the analysis of different points of the visual field, and has profound consequences for visual performance (e.g., Virsu and Rovamo, 1979; Pointer, 1986; Duncan and Boynton, 2003). Whereas the general features of the V1 map have been shown to be constant across most, if not all, mammalian species (Rosa et al., 1993), quantitative information across the entire visual field representation has become available for only a few species. Among other factors, the complex geometry of V1, particularly in the representation of the visual field periphery, creates challenges for accurate quantification. The introduction of computational and mathematical techniques for the reconstruction and visualization of cortical structure, and for the measurement of distances across curved surfaces, provides an opportunity to achieve a new level of precision in defining the visuotopic organization of V1.

Additional Supporting Information may be found in the online version of this article.

The first two authors contributed equally to this work.

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In the present study we used computational techniques to reconstruct the V1 visuotopy of the marmoset monkey (*Callithrix jacchus*), a highly visual diurnal species of New World monkey. The marmoset is one of the smallest living primates, with a brain that is less than 1/12 the volume of a macaque brain, and 1/180 the volume of the human brain (Stephan et al., 1981). There have been many studies of the organization of the visual system in the marmoset, including the retina (e.g., Troilo et al., 1993; Goodchild et al., 1996; Wilder et al., 1996; Silveira et al., 1999; Chan et al., 2001; Szmajda et al., 2005, 2008; Springer et al., 2011), lateral geniculate nucleus (LGN, e.g., Spatz, 1978; Fritschy and Garey, 1986; Yeh et al., 1995; Kremers and Weiss, 1997; Martin et al., 1997; White et al., 1998, 2001; Felisberti and Derrington, 2001; Solomon et al., 2002, 2010; Webb et al.., 2002), V1 (e.g., Fritsches and Rosa, 1996; Bourne et al., 2002; Zinke et al., 2006; Buzas et al., 2008; Yu et al., 2010), and extrastriate cortex (e.g., Krubitzer and Kaas, 1990; Rosa et al., 1997a,b, 2000, 2005, 2009; McLoughlin and Schiessl, 2006, 2009; Lui et al., 2006, 2007a,b; Federer et al., 2009; Jeffs et al., 2009, 2012; Solomon et al., 2011). The present results build on earlier reports of the global visuotopy (Fritsches and Rosa, 1996) and local precision (McLoughlin and Schiessl, 2006) of the V1 map in this species by providing comprehensive quantitative analyses. In addition, we provide the digital files underlying the present results as Supporting Materials, in a format that will allow additional analyses by other investigators.

One of the most striking features of V1 visuotopy in primates is that the representation of central vision is highly magnified. This emphasis is usually quantified using cortical magnification factors, which are ratios of the distance between two points in the cortex and the angular distance between the two points of visual field represented therein (Daniel and Whitteridge, 1961; Hubel and Wiesel, 1974; Gattass et al., 1981), or between the surface area of a region of cortex and the area of visual field sector it represents (Van Essen et al., 1984; Gattass et al., 1987); the latter is usually referred to as the areal magnification factor (M_A). According to some studies, the variation of M_A in different parts of V1 is a direct reflection of the packing density of ganglion cells in different parts of the retina (Wässle et al., 1989, 1990). However, other studies have found that V1 has a stronger bias towards representation of the central visual field than expected from retinal anatomy (Myerson et al., 1977; Schein and de Monasterio, 1987; Adams and Horton, 2003a). The first objective of this study was to accurately measure M_A in the marmoset V1 in order to provide a solid comparison with the well-characterized distribution of retinal ganglion cells in this species (Wilder et al., 1996).

Our second objective was to address the relationship between the magnification factor and receptive field size in marmoset V1. It has long been recognized that as the magnification factor decreases from the central to the peripheral visual field, there is an increase in the size of the receptive fields of V1 cells. In macaques, Hubel and Wiesel (1974) proposed that the functions relating magnification factor and receptive field size to receptive field eccentricity are inversely proportional, in a manner that results in each point of the visual field being represented by a constant number of V1 neurons. This relationship was questioned by subsequent electrophysiological studies in both New World and Old World monkeys (Dow et al., 1981; Van Essen et al., 1984; Gattass et al., 1987), but has more recently been supported by studies using functional imaging methods (Harvey and Dumoulin, 2011; Palmer et al., 2012).

We also investigated the degree of anisotropy in the V1 map of the marmoset, i.e., whether linear estimates of the magnification factor depend on the direction chosen for the measurement (Van Essen et al., 1984). While non-uniformities in linear magnification may not necessarily affect M_A, they may be indicative of some other feature of V1 architecture. In macaques and capuchin monkeys, it has been proposed that the V1 map is distorted by the pattern of ocular dominance columns (Le Vay et al., 1985; Tootell et al., 1988; Rosa et al., 1992; Blasdel and Campbell, 2001). According to this argument, the need to duplicate the representation of each point of the visual field, once for each eye, would result in the V1 map becoming "stretched" in the direction perpendicular to the preferred orientation of the ocular dominance stripes (Sakitt, 1982). Ocular dominance columns are either absent (Spatz, 1989; Roe et al., 2005) or poorly developed (Markstahler et al., 1998; Chappert-Piquemal et al., 2001) in adult marmosets, raising the question of whether or not anisotropy is present. Finally, the accurate reconstruction provided by the present results also made it possible to examine other outstanding questions about the exact configuration of the V1 map, such as whether or not the quantitative characteristics of the map vary according to angular position in the visual field, including between points near the horizontal versus vertical meridians of the visual field, or between the representations of the upper and lower quadrant (Van Essen et al., 1984; Schira et al., 2007).

MATERIALS AND METHODS

The focus of the present report is on data obtained from two adult common marmosets (*Callithrix jacchus*), which had electrophysiological recordings covering all of V1, as well as adjacent areas (results to be reported separately). Receptive fields and parasagittal sections from one of these cases (MK9) have been illustrated in a previous article (Fritsches and Rosa, 1996). Acute experiments were conducted in accordance with the Australian Code of Practice for the Care and Use of Animals for Scientific Purposes, and all procedures were approved by the Monash University Animal Ethics Experimentation Committee.

Preparation

The preparation for single-unit recordings followed the general protocol described by Bourne and Rosa (2003), with slight changes as indicated below. The animals were premedicated with intramuscular injections of diazepam (3 mg/kg) and atropine (0.2 mg/kg) and, after 30 minutes, were anesthetized with either ketamine (20 mg/kg) and xylazine (3 mg/kg), in case MK9, or alfaxalone (Alfaxan, 10 mg/kg), in case CJ119. A tracheotomy was performed, the femoral vein was cannulated, and a craniotomy was performed from the occipital pole to the approximate anteroposterior level of the rostral tip of the calcarine sulcus (AP 0; Paxinos et al., 2012). After all surgical procedures were completed the animal was administered an intravenous infusion of pancuronium bromide (0.1 mg/kg/h) combined with sufentanil (6–8 μg/kg/h) and dexamethasone (0.4 mg/kg/h), after which it was artificially ventilated with a gaseous mixture of nitrous oxide and oxygen (7:3). The electrocardiogram and SpO_2 level were continuously monitored. Administration of atropine (1%) and phenylephrine hydrochloride (10%) eye drops resulted in mydriasis and cycloplegia. Appropriate focus and protection of the cornea were achieved by means of polymethylmethacrylate contact lenses fitted with carmellose sodium eye drops. Visual stimuli were presented monocularly to the eye contralateral to the hemisphere from which the neuronal recordings were obtained. The ipsilateral eye was occluded.

Recording

Parylene-coated tungsten microelectrodes ($\sim$1 MΩ) with exposed tips of 10 μm were directed towards V1. The electrode was slowly advanced into the cortex until the first units could be observed above background activity, following which excitatory receptive fields were mapped at 300–500-μm intervals. Visual stimulation was achieved through a manually operated light source, which was projected onto a translucent hemisphere (e.g., Yu and Rosa, 2010), by correlating the stimulation of specific portions of the visual field with increments in the neural activity. Visual stimuli comprised luminous white spots (1–10° in diameter), which were used to explore the approximate locations of the receptive fields, and bars (2–20° long, 0.2–1° wide), which were used to carefully map the boundaries of the receptive fields, taking into consideration the neuronal selectivity for orientation. The receptive fields were drawn as rectangles parallel to the axis of best orientation, or, in the rare cases where an orientation bias was not obvious, as ovals. Small electrolytic lesions (4 μA, 10 sec) were placed at various sites during the experiments to aid in the histological reconstruction of the recording sites.

Histology

At the end of the experiments the animals were administered an overdose of sodium pentobarbitone and perfused transcardially with 0.9% saline, followed by 4% paraformaldehyde in 0.1M phosphate buffer (pH 7.4). After cryoprotection by increasing concentrations of sucrose, the brains were sectioned in the parasagittal (MK9) or coronal (CJ119) planes using a cryostat. To aid in the alignment of sections during computerized reconstruction of V1, reference pinholes were created perpendicular to the intended plane of sectioning by inserting a three-pronged set of pins through the fixed brain prior to the cryoprotection steps. The known separation of the pins was used to estimate shrinkage due to subsequent histological processing, allowing correction of our quantitative estimates. A series of slides (40 μm) was stained for Nissl substance; interleaving sections were stained for myelin (both animals) and, in case CJ119, cytochrome oxidase (Wong-Riley, 1979).

Reconstruction and analysis

Cytochrome oxidase and Nissl-stained sections were scanned with a digital flatbed scanner and then aligned using the pinholes created prior to sectioning. Layer 4 was traced in each section using the program ImageJ (Abramoff et al., 2004), 3D surface models of the entire cortex were reconstructed with Caret (Van Essen et al., 2001), and irregularities in the mesh, introduced due to slight irregularities during mounting of the sections, were smoothed using MeshLab (Cignoni et al., 2008). The boundaries of V1 were determined histologically (Paxinos et al., 2012), following which the 3D surface of this area was cut out of the whole brain model using the Dijkstra (1959) algorithm to find the shortest path between boundary points.

Electrode recording sites were reconstructed using the tracks left in histological sections, and electrolytic lesions, as well as the depth notes recorded during the experiment. Each recording site was then computationally associated with the closest vertex in the V1 mesh. Visuotopic information (in the form of eccentricity and polar angle of the center of each receptive field) was interpolated across the whole mesh using the method described by Sereno et al. (1994). This interpolation procedure estimated visuotopic coordinates at all mesh nodes using a distance weighted smoothing algorithm. The horizontal meridian of the visual field was defined as a line connecting the centers of the fovea and blind spot (Troilo et al., 1993), and the vertical meridian was defined as a line perpendicular to the horizontal meridian through the center of the fovea. The estimates of the exact location of the center of the fovea were refined *a posteriori*, based on the assumption that the most central receptive fields were represented at the "apex" of V1, on the lateral surface of the occipital lobe (Dow et al., 1981; Tootell et al., 1988). The apex is easily recognizable in 2D and 3D models within 0.5 mm accuracy (Tootell et al., 1988). This step was necessary given that the precision of the method for estimating the projection of the center of the fovea onto the hemispheric screen during the experiment is $\sim 0.5^\circ$ (Gattass et al., 1987). Flat maps of V1 were produced using Caret. To accurately measure distances on the mesh we used an exact geodesic measurement (O'Rourke, 1999). Isopolar and isoeccentricity contours were determined by finding the geodesic shortest path between mesh vertices that lay on, or nearest to, the visuotopic contour. Both flat maps and 3D models were visualized with Paraview (Henderson, 2007).

Measurements of cortical magnification factor

The areal cortical magnification factor (M_A) was calculated by first dividing the V1 map into bins defined by logarithmically spaced isoeccentricity lines up to 32° (1°, 2°, 4°, 8°, 16°, 32°) and then to 50° (approximate beginning of the monocular crescent), 60° (approximate end of the region where all polar angle segments are fully represented), and 90° (approximate maximum extent of the visual field, along the horizontal meridian). Areal magnification was then calculated as the surface area (mm^2) of the 3D V1 model within an eccentricity bin divided by the corresponding area of visual space (deg^2); this ratio was designated as the M_A value at the midpoint of the eccentricity bin. Separate estimates of M_A were also obtained for sectors defined by different polar angles, in 20° bins.

The linear cortical magnification factor (M) was defined as the distance between two points in V1, divided by the angular distance between the geometric centers of the corresponding receptive fields. The linear magnification factor was specified in two directions: either along isopolar lines (M_P) or along isoeccentricity lines (M_E). Multiplying M_E and M_P is equivalent to the areal magnification, M_A. Linear magnification is known to be approximately inversely proportional to eccentricity and is typically (Schwartz, 1994) modeled with an equation of the form:

$$M(E) = \frac{k}{E + a} \tag{1}$$

where E is eccentricity (°), and k and a are scalar constants. However, as explained below, it was apparent that such an equation could not adequately describe our measurements for M_E, so an exponential term with a new parameter, b, was added to the standard magnification function:

$$M(E) = e^{b*E} * \frac{k}{E + a} \tag{2}$$

The addition of the exponential term allows the magnification to decrease faster with eccentricity than Equation 1.

We measured magnification along isoeccentricity lines (M_E) by first measuring the length of isoeccentricity lines on the cortex. The length of an isoeccentricity line on the cortex (mm) is equal to the magnification (mm/°) multiplied by the length of the isoeccentricity line in visual space (°), allowing the fitting of a magnification function (Eqs. 1 or 2) to the distance measurements.

Quantifying magnification directly along isopolar lines (M_P) at a specific eccentricity can be problematic because magnification changes with eccentricity, and isopolar lines by definition span a range of eccentricities. One solution is to measure very short segments of isopolar lines and to attribute the obtained values of M_P to the mid-point eccentricity. However, this approach is very prone to error given local irregularities that can emerge in the interpolated maps, for example, as a result of inadequate sampling in specific regions. We avoided this problem by calculating the rate of change in isopolar line length directly on the 3D model. In other words, M_P was estimated as the derivative of the length of isopolar line, with respect to eccentricity. The length of isopolar lines increased logarithmically with eccentricity, so the dataset was fit with a log function of the form:

$$L_p = k * Log(E + a) + c \tag{3}$$

where L_P is the length of an isopolar line at eccentricity E. The magnification is calculated as the derivative of this equation, which has the same form as the standard equation (Eq. 1).

Other functions (V1 magnification factors from other species, and the marmoset ganglion cell density, as a function of eccentricity), obtained from previous studies, were converted to percentages by dividing by the total surface area (or number of cells). This was calculated by calculating the integral of the magnification function multiplied by the length of the isoeccentricity line from 0.1° eccentricity (to avoid numerical errors) to 90°. Receptive field size was measured by calculating the area (deg^2) of the receptive field polygons on the sphere.

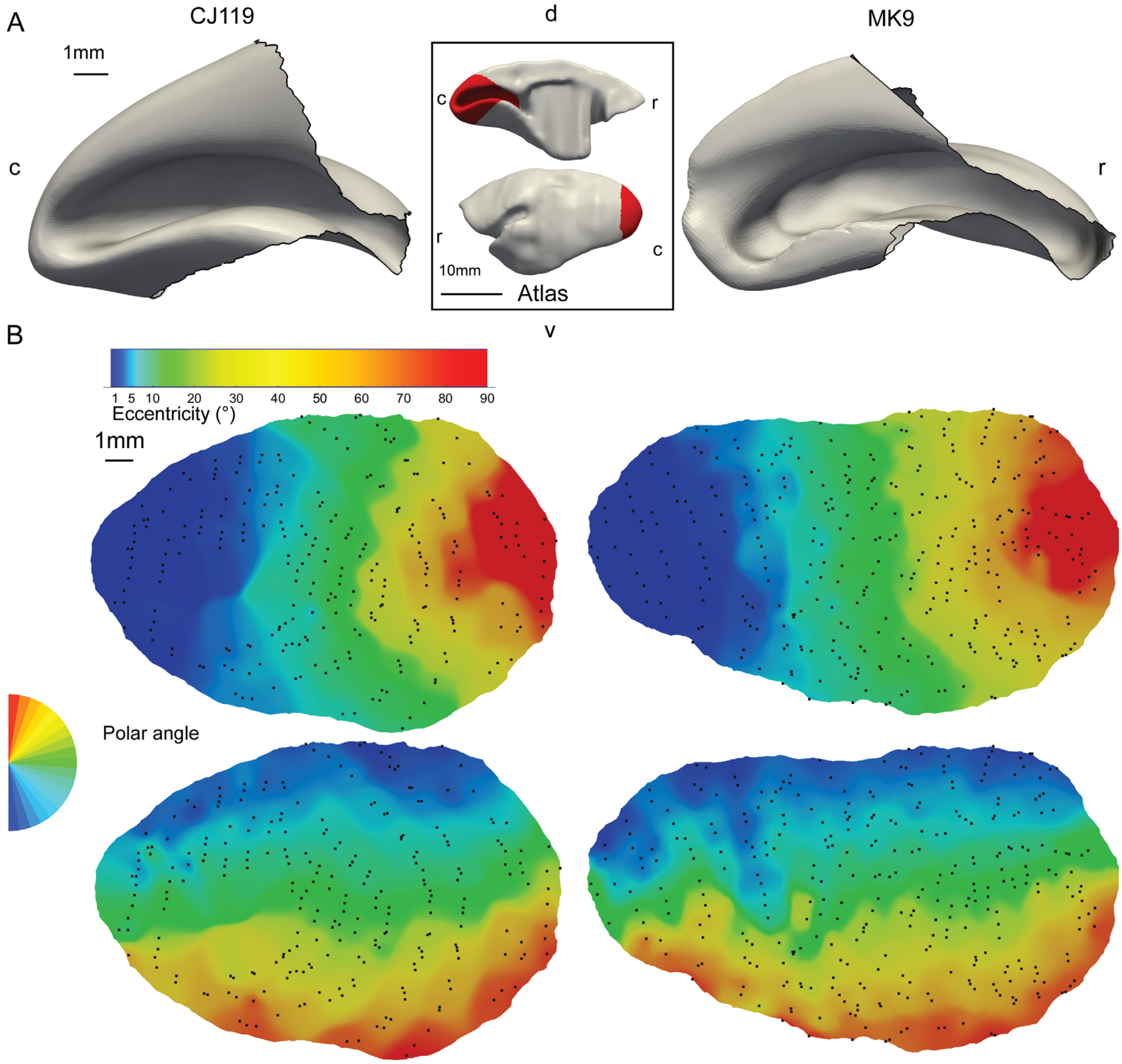

Figure 1. 3D reconstructions of V1 surfaces and receptive field maps in the two cases analyzed in this study (CJ119 and MK9). **A:** Medial views of the V1 surfaces for CJ119 (left) and MK9 (right), reconstructed from layer 4 in the left hemispheres. The inset shows the location of V1 (red) in medial (top) and caudolateral (bottom) views of the whole marmoset brain. The model was reconstructed from a publicly available atlas (Paxinos et al., 2012). c, caudal; d, dorsal; l, lateral; r, rostral. The location of the recording sites in each case is shown in Supporting Movies S1 and S2. **B:** Flattened models of both cases, showing interpolated eccentricity (top, log scale) and polar angle maps (bottom). Black dots represent recording sites, projected to the 2D map. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

RESULTS

Accurate 3D models were successfully constructed for two cases from which several hundred electrophysiological recordings were obtained, sampling almost all of V1 (Fig. 1A,B; see also Supporting Movies S1 and S2), and therefore most of the contralateral hemifield (Fig. 2). The total surface area of V1 was 193 mm^2 in case MK9, and 214 mm^2 in case CJ119. These values are similar to those reported by Fritsches and Rosa (1996; 205 mm^2), Pessoa et al. (1992; 182 mm^2), and Missler et al. (1993; 194 mm^2) in previous studies of marmoset V1. The two V1 models were slightly different in aspect ratio: case CJ119 was shorter along the rostral-caudal and dorsal-ventral axes, but wider along the lateral-medial axis, compared to case MK9. Whereas flat map representations (see Supporting Movies S3 and S4) are shown here to allow convenient visualization of the visuotopy of V1, these were found to introduce significant distortions,

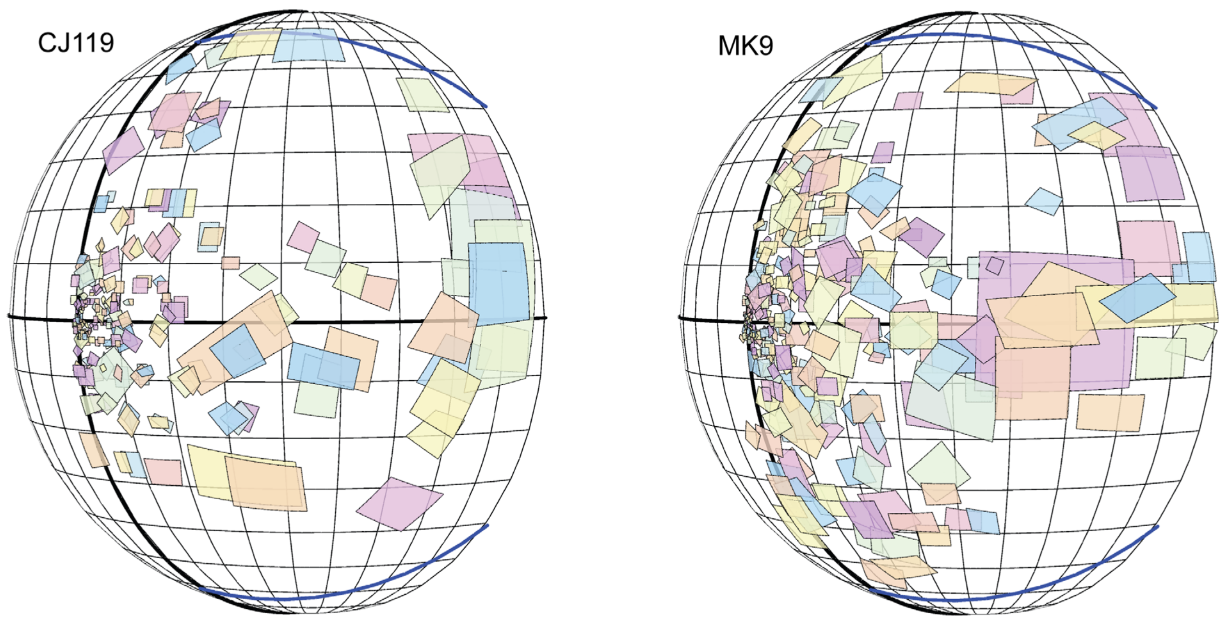

Figure 2. Plots of mapped receptive fields in CJ119 (left) and MK9 (right). Colors are assigned arbitrarily for visualization. The thicker black lines represent the horizontal and vertical meridians of the visual field, and the blue lines represent the upper and lower extent of the marmoset's visual field along the periphery (Fritsches and Rosa, 1996). Finer lines represent 10° intervals. After 60° eccentricity, some portions of the upper and lower field near the vertical meridian have no representation in V1. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

typically in the form of compression near the center of the map and expansion around the edges (Fig. 3). Thus, all measurements were performed on the 3D models. Both cases showed the stereotypical shape (Polimeni et al., 2006) and topographic organization (Fritsches and Rosa, 1996) of simian V1, with high magnification of the central visual representation (Fig. 1). As detailed in Table 1, in both animals the central 5° and 10° of visual space accounted for ~30% and 60% of the cortex, respectively, and the upper and lower field representations occupied approximately equal areas.

Areal cortical magnification factor

Estimates of the areal cortical magnification factor (M_A) are illustrated in Figure 4A. Cortical magnification was approximately inversely proportional to eccentricity and spans values from nearly 10 mm^2/deg^2, in the representation of the center of the fovea, to less than 0.01 mm^2/deg^2, in the representation of the far periphery. Magnification estimates in the two cases were very similar (Fig. 4, left and right columns).

To test if M_A varies according to the representation of polar angles, we measured the cortical areas associated with visual field segments encompassing 20° polar angle sectors. The visual field of the marmoset, like that of most monkeys, is more extensive along the horizontal meridian than along the vertical meridian (the blue lines in Fig. 2 represent the limits of the field of vision), and the representation in V1 follows a similar pattern (Gattass et al., 1987; Fritsches and Rosa, 1996). In the far periphery (greater than 60° eccentricity), regions of the upper field and lower field near the vertical meridian are not represented in V1, whereas the area near the horizontal meridian is (Figs. 2, 4C). Thus, we present our polar angle area measurements in two groupings: below 60° eccentricity, and above 60°. For the area representing eccentricities below 60°, cortical area was approximately equal for each segment (Fig. 4B), except in the 20° segments immediately adjacent to the vertical meridian. This was likely due to sampling and the interpolation of the map—if there are no cells sampled next to the border between V1 and the second visual area (V2), then the interpolation method will map the nearest recording site, thereby underestimating this part of the visual field. For the polar segments in the region greater than 60° eccentricity (Fig. 4C), the area of cortex was approximately what was expected if magnification is unaffected by polar angle, but suffered from a similar effect near the vertical meridians; given the small area of this region (less than 8 mm^2), sampling errors were likely to have a greater effect. In summary, there was no clear evidence of a systematic relationship between polar angle and areal magnification.

Linear cortical magnification factor

Linear cortical magnification was measured in two directions: along isoeccentricity lines (M_E) and along

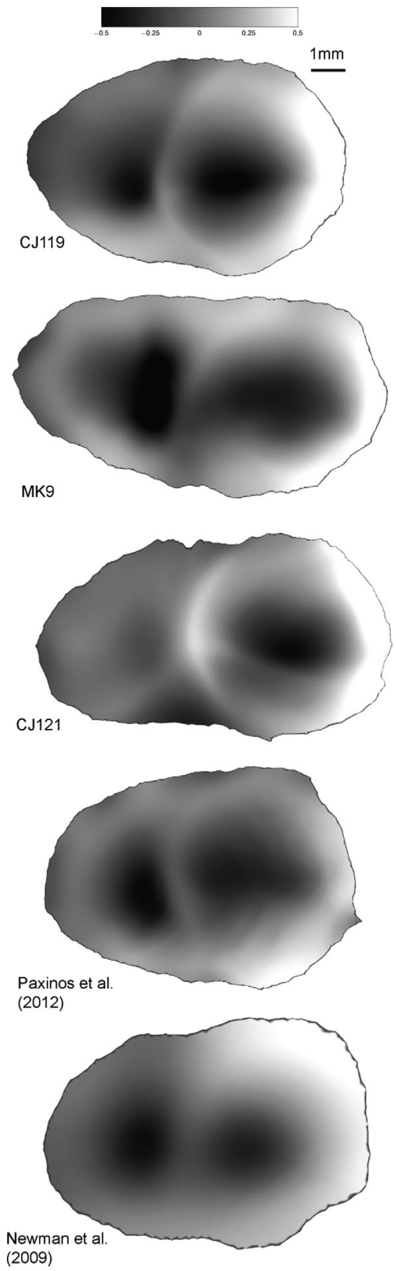

Figure 3. 2D reconstructions of V1 in five marmosets, showing the patterns of areal distortion incurred in forcing a 2D solution to the shape of V1. These maps were prepared with the program Caret (Van Essen et al., 2001). As indicated in the scale bar (top), darker colors indicate areal compression (black corresponding to 50% compression in area), and lighter colors indicate expansion (white corresponding to 50% expansion). The cases include three animals from studies conducted in our laboratory (CJ119, MK9, and CJ121), as well as two from publicly available digital atlases of the marmoset brain (Newman et al., 2009; Paxinos et al., 2012). Animations showing the process of converting the 3D shape of V1 into flat maps are provided as Supporting Movies (S3 and S4).

isopolar lines (M_P). We estimated magnification along isoeccentricity lines by measuring the lengths of isoeccentricity lines (Fig. 5A) and dividing these by the lengths of

TABLE 1.
Sample Characteristics and Comparison Between Cases

Case	CJ119	MK9
Sectioning plane	Coronal	Parasagittal
Number of receptive fields recorded in V1	264	333
Rostral-caudal length (mm)	9.5	12.0
Medial-lateral width (mm)	8.5	7.7
Dorsal-ventral height (mm)	4.5	6.3
Surface area (mm^2)	193	214
Cortical area of upper field representation (%)	53	53
Cortical area representing eccentricities $\leq 5°$ (%)	32	30
Cortical area representing eccentricities $\leq 10°$ (%)	63	60

the corresponding isoeccentricity contours in the visual field. The isoeccentricity line lengths increased with eccentricity until $\sim 8°$, beyond which they decreased, as expected by the approximately elliptical shape of V1 (Figs. 1, 3, 5A). Using Equation 2, magnification functions were fitted to these measurements, resulting in the following functions (Fig. 5C, continuous lines):

$$\text{CJ119}: \quad M_E(E) = e^{-0.0065E} * \frac{5.52}{E + 0.71} \tag{4}$$

$$\text{MK9}: \quad M_E(E) = e^{-0.0052E} * \frac{4.82}{E + 0.61} \tag{5}$$

Magnification along isopolar lines (M_P) was estimated by calculating the rate of change (derivative) of isopolar line length (Fig. 5B). The magnification factors along isopolar lines, calculated using this method (Fig. 5C, dashed lines), were:

$$\text{CJ119}: \quad M_p(E) = \frac{3.73}{E + 0.26} \tag{6}$$

$$\text{MK9}: \quad M_p(E) = \frac{4.91}{E + 0.56} \tag{7}$$

As shown in Figure 5B, there was no clear relationship between polar angle and isopolar line length in CJ119. In MK9, isopolar lines near the vertical meridian (red symbols) seemed to be slightly shorter than those near the horizontal meridian and intermediate polar angles. This is the opposite of what is predicted by flat V1 models (Schwartz, 1994; Balbasumarian et al., 2002; Schira et al., 2007, 2010) but is compatible with a curved model (Rovamo and Virsu, 1984).

The product of M_P and M_E is equivalent to the areal magnification function (M_A). In Figure 4A we compare the function obtained by the product of these measurements with our estimates of areal magnification. In both cases it is clear that the results of these two analyses match very closely.

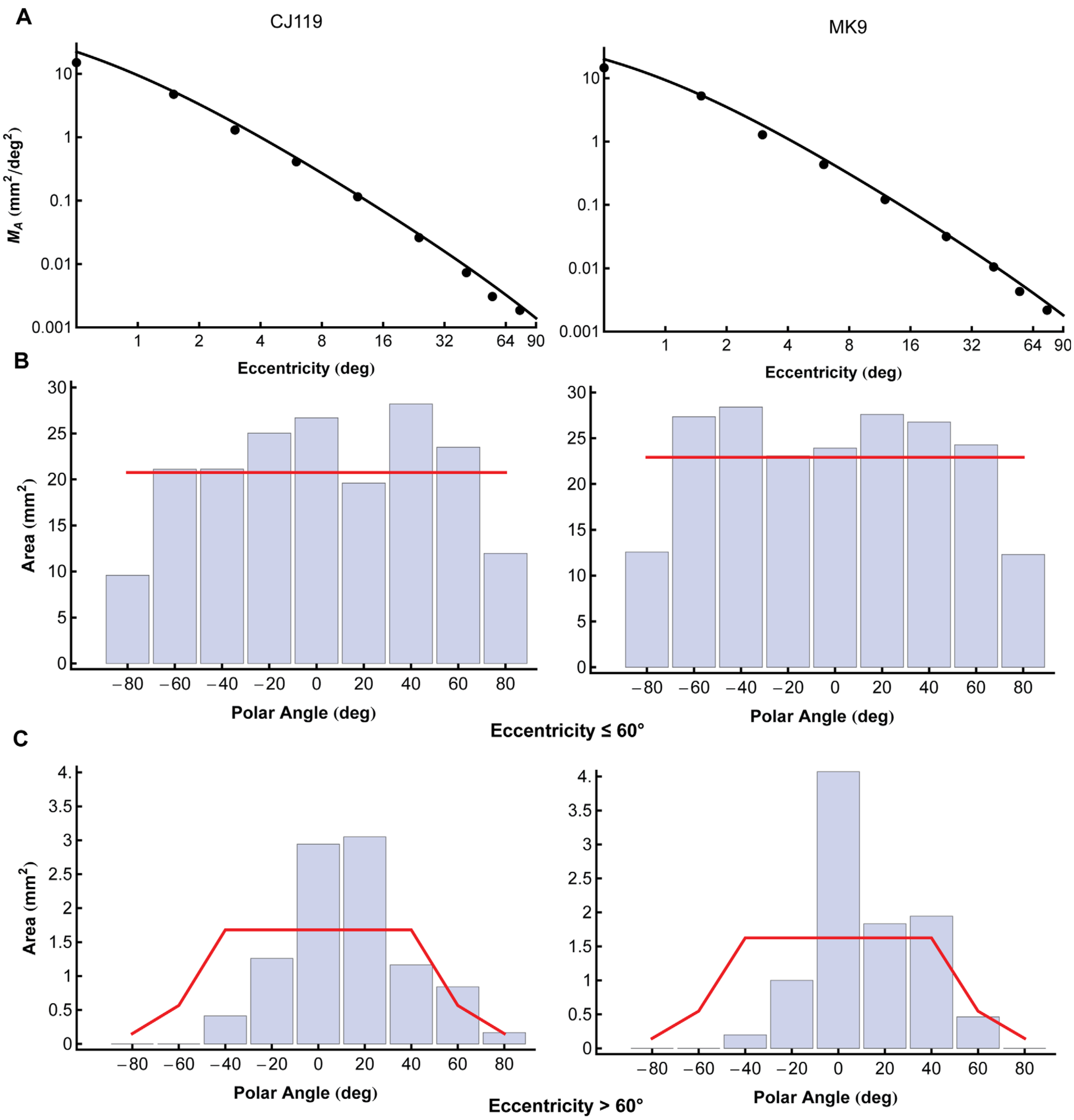

Figure 4. Areal cortical magnification in two cases, as a function of eccentricity and polar angle. **A:** Areal magnification (M_A) plotted on a log-log scale. The circles represent the areal magnification estimated by binning the cortex by eccentricity ranges, and dividing the resulting cortical surface areas (mm^2) by the areas of the corresponding regions of visual space (deg^2). The plotted line is the product of the linear magnification functions measured in the direction of isoeccentricity lines (M_E) and isopolar lines (M_P). There is a very good correspondence between these estimates. **B:** Area of cortex associated with polar angle segments of 20° in the first 60° of eccentricity. The red curve represents the expected area if the polar angle does not affect magnification, i.e., equal visual segments have equal cortical areas. **C:** Area of cortex associated with polar angle segments of 20° beyond 60° of eccentricity. In this region not all polar angle segments are fully represented in V1 (see Fig. 1C); the areas near the vertical meridian have little or no representation. The red curve represents the approximate expected areas under the assumption that polar angle does not affect cortical magnification. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

Next we investigated the relationship between M_E and M_P. If M_E and M_P are equal at a given point in space, then magnification is *isotropic*; otherwise it is *anisotropic*. Anisotropy (A) can be quantified as the ratio of M_P to M_E (Van Essen et al., 1984):

$$A(E) = \frac{M_F(E)}{M_E(E)} \tag{8}$$

In both cases M_E decreased more rapidly than M_P in the periphery (Fig. 5D). In CJ119 (purple curve) the values of

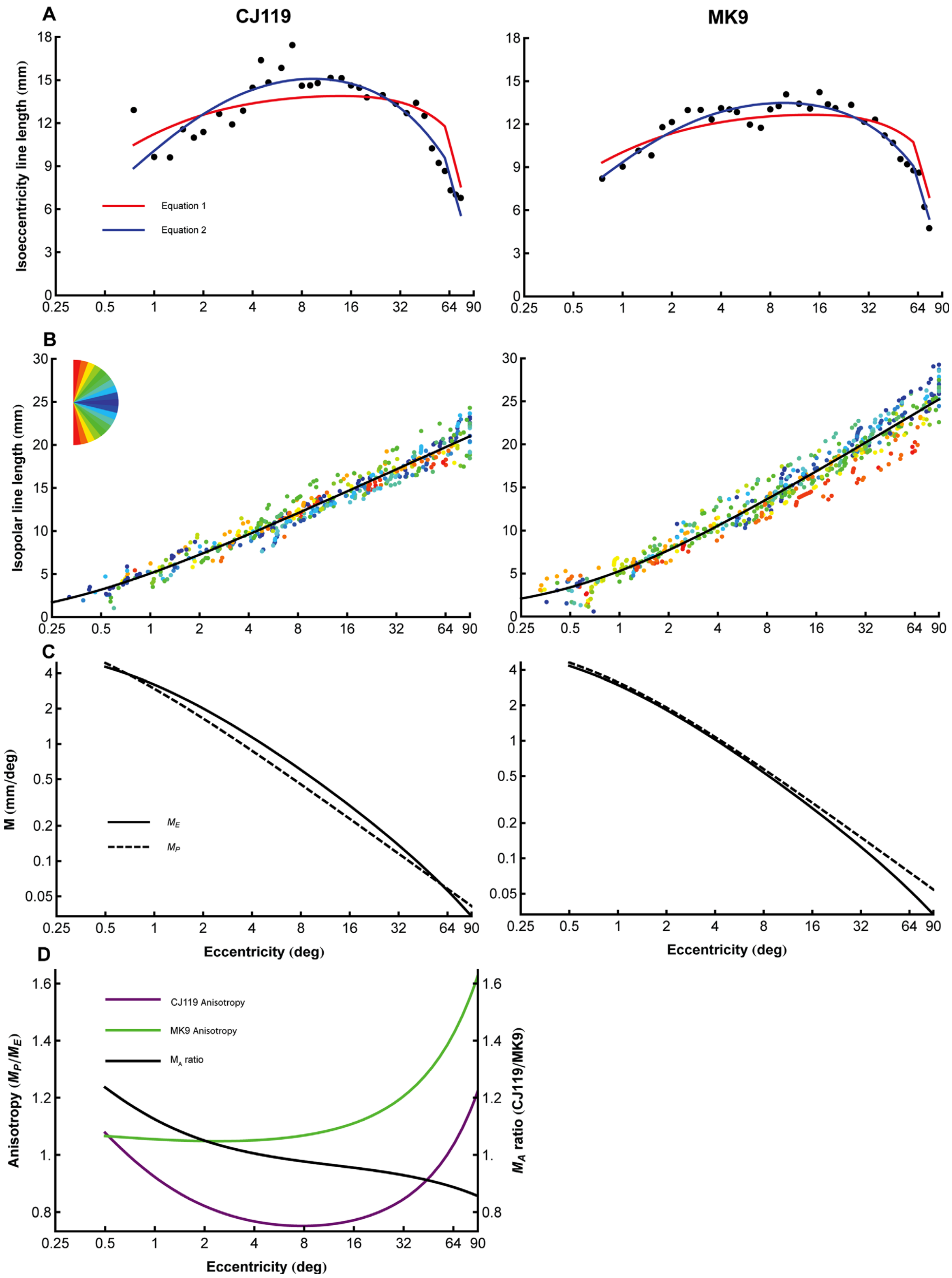

Figure 5. Calculation of linear cortical magnification functions and anisotropy. **A:** The linear cortical magnification function along isoeccentricity lines (M_E) was estimated by fitting the length of isoeccentricity lines to Equations 1 and 2 (see Materials and Methods). Here the length of isoeccentricity lines (black circles) at varying eccentricities is plotted on a semi-log plot, with the best-fitting lines provided by Equation 1 (red) and Equation 2 (blue). Equation 2 provided a much better fit, capturing the sharp rise and fall in length as a function of eccentricity. **B:** Isopolar line length (colored dots) plotted on a semi-log plot against eccentricity. Individual measurements are color-coded according to polar angle (inset), with the logarithmic line of best fit shown by the black line (Eq. 3). **C:** The resulting M_E (solid line) and M_P (dotted line) functions plotted for both cases. **D:** Ratio of the values of M_P and M_E functions, using the anisotropy index (Eq. 4) across eccentricities in CJ119 (purple line) and MK9 (green line). Both cases show some degree of anisotropy but the amount and distribution is somewhat different. Also shown is the ratio between the M_A functions of the two cases (black line). Despite differences in anisotropy, areal magnification is very similar in the two cases (ratio near 1 across eccentricities). [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

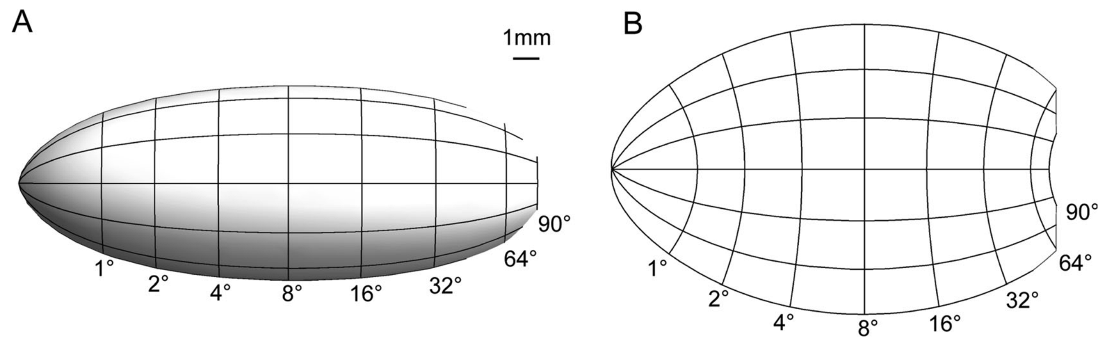

Figure 6. Summary map of marmoset V1. **A:** The magnification function (derived from Eq. 9) is visualized using the surface-of-revolution model proposed by Rovamo and Virsu (1984). **B:** The same function visualized using the flat model proposed by Schira et al. (2007) (parameters used k = 4.79, a = 0.86, b = 73.2, alpha = 1.09). Note that the curved surface in (A) creates an artificial impression that V1 is long and narrow in 2D view. The true aspect ratio of V1 is more readily appreciated in (B).

M_P and M_E were within 20% of each other across the entire range of eccentricities, with M_E slightly exceeding in M_P in the first 50°, and M_P exceeding M_E beyond this point. In contrast, V1 in case MK9 (red curve) was nearly isotropic for the first 30° of eccentricity (M_P and M_E showing no more than a 10% difference), beyond which M_P outweighed M_E. These findings are compatible with the appearance of the 3D models and flat maps: in case CJ119 V1 was relatively compact, whereas in MK9 it was more elongated, reflecting the shape of a relatively long calcarine sulcus (Figs. 1A,B, 2; Table 1). Despite the variability in the shape and anisotropy of V1 the ratio of the areal magnification factors in the two cases (black line, Fig. 5D) was very similar: M_A differed by no more than 20%, and these extremes corresponded to very central and peripheral parts of V1 (regions which are more prone to measurement errors). Between the more reliably measured eccentricity range of 2 to 50°, the difference in magnification between cases was less than 5%.

Equation 9, based on parameters derived from Equations 4–7, scaled to match the measured surface area of V1, was used to summarize the visuotopy of marmoset V1.

$$M_A(E) = \left(\frac{4.32}{E + 0.53}\right)^2 * e^{-0.0059*E} \qquad (9)$$

The magnification function is visualized in Figure 6 using two mathematical models. The first model (Fig. 6A) is based on Rovamo and Virsu (1984). It is a model with intrinsic curvature and it is consistent with our observation that isopolar lines have similar lengths. The second model (Fig. 6B) is based on a complex log function (Schira et al., 2007), which was generated using the method described by Polimeni et al. (2006). It is intrinsically flat and its predicted isopolar line lengths deviate from our data, but is easier to visualize on a 2D plane.

Comparison with other species of primate

Having calculated a marmoset cortical magnification function in terms of percentage area of the surface area of V1, we could compare this to magnification functions obtained from other primate species. Figure 7A,B shows the functions derived for marmosets (this study), bushbabies (Rosa et al., 1997a,b), squirrel monkeys (Adams and Horton, 2003a), macaques (Van Essen et al., 1984), and humans (Schira et al., 2007). It is clear that all diurnal species have similar magnification factors despite differences in size, with the possible exception of the squirrel monkey, which appears to have slightly lower magnification in the central and far peripheral visual field (although to some extent this may reflect distortions introduced during the process of physical flat-mounting; Adams and Horton, 2003a). In comparison, the function obtained for the nocturnal bushbaby, *Galago*, deviates significantly, with a relatively lower magnification of the representation of central vision.

Comparison with the retinal ganglion cell density

Using Equation 9, we compared the visuotopic map of marmoset V1 to the retinal ganglion cell density function from Wilder et al. (1996), as well as LGN magnification data from White et al. (1998). All magnification functions were scaled to be in terms of percentage (% V1 surface area, % LGN volume, and % of ganglion cells) per deg^2. It is clear from Figure 7C that the relative magnification of central vision in V1 is much higher than that predicted by the variation of ganglion cell density (see also Table 2). For example, the central 2° of visual space occupies 21.4% of the surface of V1, but only project to 7.1% of the ganglion cells. LGN magnification data are more limited (only four data points were available) so it is not clear whether it is more similar to that of V1, the retinal

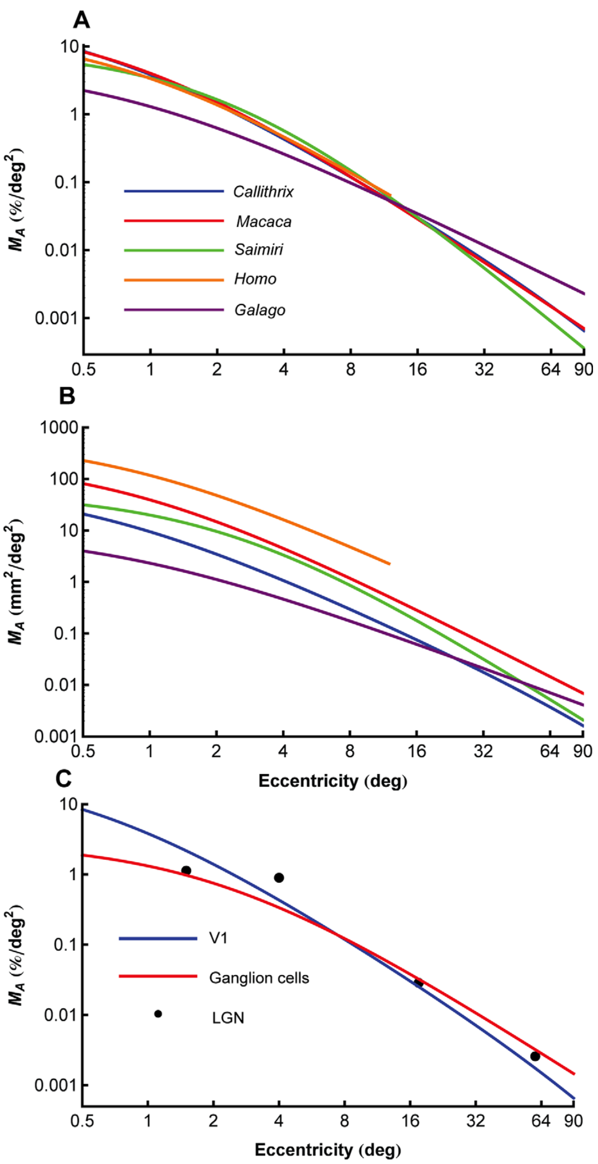

Figure 7. Comparisons with other species of primate, and with the retinal ganglion cell density of the marmoset. **A:** Comparison of the relative V1 areal magnification as a function of eccentricity (expressed in percentage of V1 area per deg^2). Using this format, the marmoset curve (blue) is obscured by the function derived for the macaque (red). Functions derived for the squirrel monkey (green), and humans (orange, central 12° only) follow very similar trajectories. In contrast, the function for the nocturnal *Galago* deviates from those obtained for the diurnal species, with a relatively lower emphasis on central vision. **B:** Comparison of V1 areal magnification for the same species shown in (A), but expressed in raw linear magnification values (mm^2/deg^2). **C:** V1 areal magnification factor in the marmoset (blue) compared with ganglion cell density (red; from Wilder et al., 1996) and LGN magnification (black circles; from White et al., 1998). All three types of magnification are expressed as percentages (% V1 surface area, % ganglion cells, % LGN volume). V1 magnification is clearly much higher in the central 4° (see Table 2 for numeric values). [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

TABLE 2.
Percentage of Retinal Ganglion Cells and V1 Neurons Representing Different Eccentricity Ranges

Eccentricity	Ganglion cells (%)	V1 surface area (%)
0–1°	2.6	11.1
1–2°	4.5	10.3
2–4°	9.1	13.7
4–8°	14.4	15.9
8–16°	19.1	16.7
16–32°	21.9	16.0
32–64°	21.9	13.1
64–90°	6.5	3.2

ganglion cell density, or lies somewhere in between (Malpeli and Baker, 1975).

Point-image size

The product of the magnification factor and receptive field size yields an estimate of the amount of cortex activated by a single point of light, which has been termed the point image size (Capuano and McIlwain, 1981; Van Essen et al., 1984). A determination of the full extent of the point image size requires knowledge of receptive field scatter along a cortical column (Hubel and Wiesel, 1974; Albright and Desimone, 1987). However, if it is assumed that there is a constant relationship between the sizes of individual receptive fields and scatter throughout V1 (Hubel and Wiesel, 1974; Van Essen et al., 1984; Gattass et al., 1987), then the product of receptive field area and M_A will give a minimum estimate of point image size (PI). We found that PI varied from ~5 mm^2 in the foveal representation, to less than 0.2 mm^2 in the periphery (Fig. 8A). This relationship was highly significant, and was very similar in the two animals:

$$\text{CJ119}: \mathrm{PI(E)} = 2.61 * E^{-0.74}\ (r^2 = 0.47,\ P < 0.01) \tag{10}$$

$$\text{MK9}: \mathrm{PI(E)} = 2.88 * E^{-0.65}\ (r^2 = 0.45,\ P < 0.01) \tag{11}$$

Relationship between V1 receptive fields and retinal ganglion cell density

Finally, we explored if there is a relationship between the sizes of V1 receptive fields and the density of retinal ganglion cells in one hemiretina, which was assessed based on the following equation given by Wilder et al. (1996):

$$\text{Ganglion cells/deg}^2 = 5.552 * 10^4 * (E + 2.06)^{-2}$$

The results of this analysis (Fig. 8B) revealed that the number of ganglion cells contained in the retinal area equivalent to one V1 receptive field is approximately

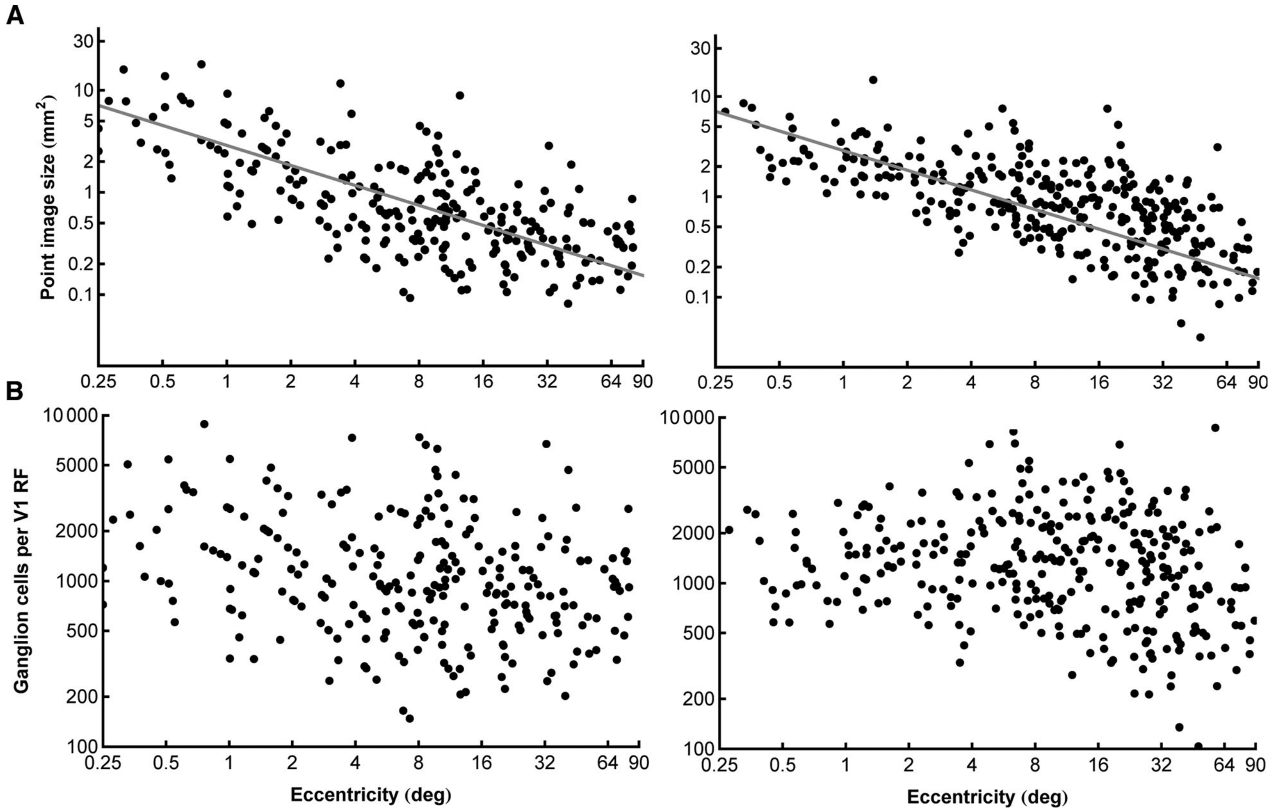

Figure 8. Analyses based on the receptive field size of V1 cells. A: The point image size, estimated by the product of receptive field area and M_A, is plotted on a log-log scale for two animals. The best-fitting power function for each case is illustrated. B: Number of retinal ganglion cells per area of visual field corresponding to one V1 receptive field, plotted on a log-log scale. No significant trend was observed in either case. RF, receptive field.

constant throughout the visual field (Pearson's correlation coefficient CJ119: $r^2 = 0.02$, MK9: $r^2 = 0.02$). Thus, despite the discrepancy in the relative sampling densities of central and peripheral space in the retina and cortex (Fig. 7C), V1 cells with receptive fields across most of the visual field receive information from a similar number of retinal ganglion cells. Combining data from both cases, there were on average 1,659 ganglion cells (from each eye) per V1 receptive field (SEM = 71, $n = 597$). Particularly in case MK9 (Fig. 8B, right), there was a suggestion that this relationship was changed in the representation of the far periphery ($>20°$ eccentricity).

DISCUSSION

We have quantified the areal and linear cortical magnification factors, and established the relationship between these variables and receptive field size, based on extensive electrophysiological recordings in the marmoset V1. Whereas the scope of our dataset is comparable to those forming the bases of previous electrophysiological studies of macaques (Van Essen et al., 1984) and capuchin monkeys (Gattass et al., 1987), the present study took advantage of more recent developments in techniques for graphic reconstruction and mathematical analysis to achieve a more precise quantification of the visual topography of V1. Recent studies based on high-resolution functional magnetic resonance imaging (MRI) have used similar computational tools to reconstruct the representation of the central visual field in the human brain (Schira et al., 2007, 2009), in this way reaching a level of precision comparable to that achieved by electrophysiological and metabolic mapping studies of the same portion of V1 in the macaque monkey (Dow et al., 1985; Tootell et al., 1988). In addition to its comparative value, the present work builds on these studies by extending the quantification of the main features of the V1 map to the peripheral visual field. In the squirrel monkey, Adams and Horton (2003a) also obtained a comprehensive view of the V1 map, based on variations in the reactivity of cortical tissue to cytochrome oxidase, following monocular enucleation. The present methods for computational reconstruction circumvent the possibility of imprecisions derived from the physical flat mounting of the cortex, which introduces significant distortions to the shape and size of certain parts of V1 (Fig. 2).

Relationship between cortical magnification factor and polar angle

Our analyses demonstrate that the areal magnification factor (M_A) in V1 is primarily defined by the receptive field eccentricity, but is not affected by polar angle. The results in this respect were very similar in the two animals studied. However, results for linear magnification factor and anisotropy varied between cases. In case CJ119 there was no effect of polar angle on estimates of M_P or M_E. In contrast, in case MK9 M_P was lower along the representation of the vertical meridian, compared to that of the horizontal meridian. As M_A was not affected by polar angle, M_E increased in this region in order to compensate. These results are important in that they indicate a developmental mechanism that ensures nearly equal representation of all polar angle segments of the retina in V1, at a given visual field eccentricity. However, they also indicate that this end result can be achieved in different ways, in terms of the detailed structure of the visuotopic map. In other words, M_A is preserved even when the map is stretched or compressed along specific axes, in different individuals. This interpretation is consistent with the findings of Adams and Horton (2003a) in the squirrel monkey and Schira et al. (2007) in humans. Comparison of the two animals described here with a larger sample (Fig. 2) indicates that there is significant variability in the shape of V1, with MK9 having one of the most elongated maps among the cases we have studied so far.

Studies of humans (Schira et al., 2007) and squirrel monkeys (Adams and Horton, 2003a) have described larger effects of visual field polar angle on linear magnification factors of V1, typically in terms of M_P being increased near the vertical meridian, with a concomitant decrease in M_E. In macaques and capuchin monkeys this has been linked to the pattern of ocular dominance stripes being typically aligned perpendicular to the V1/V2 border, but more randomly oriented along the representation of the horizontal meridian (Sakitt, 1982; Gattass et al., 1987; Tootell et al., 1988; Rosa et al., 1992; Blasdel and Campbell, 2001). Our measurements of magnification factor in the marmoset do not accord with this pattern, perhaps reflecting the fact that ocular dominance columns are indistinct in this species (Markstahler et al., 1998; Chappert-Piquemal et al., 2001). In the squirrel monkey, another species of small, diurnal New World monkey, Adams and Horton (2003b) demonstrated that the pattern of ocular dominance columns is highly variable, and it is possible that variations in the degree of anisotropy in different marmosets are linked to the degree of definition and local topography of ocular dominance stripes. However, the cause–effect relationship underlying this observation remains unclear, i.e., whether the need to accommodate segregated representations of the two eyes according to an oriented pattern of stripes results in distortions in the shape and topography of V1 during development, or if the pattern of stripes develops interactively given a predetermined shape of V1, in order to optimize coverage while maintaining a continuous representation of the visual field (Goodhill, 1993). Furthermore, mathematical models of the shape and visuotopy of V1 based on the log polar transform all predict that the representation of the vertical meridian is $\sim$1.4 times longer than that of the horizontal meridian (Schwartz, 1984; Balasubramanian et al., 2002; Schira et al., 2010). Given that no flat model of V1 can have equal length isopolar lines along the different meridians (as observed in the marmoset), it is clear that V1 in this species is intrinsically curved (Rovamo and Virsu, 1984; Motter and Simoni, 2007). This conclusion is compatible with our measurements of the distortions introduced by the process of forcing a flat solution to representing the shape of the marmoset V1 (Fig. 2). However, there is evidence that the human V1 is intrinsically flat, at least in the central representation (Schira et al., 2007). Hence, it is possible that the intrinsic curvature of V1 is also determined by other requirements of the cortical architecture.

Magnification in the periphery

We found that linear magnification factor along isoeccentricity lines (M_E) decreases faster with eccentricity than the prediction based on Equation 1. In all three species of monkey for which detailed maps of V1 have been obtained, the function describing the variation of M_E with eccentricity was found to decrease faster than that describing the variation of M_P (macaque: Van Essen et al., 1984; capuchin monkey: Gattass et al., 1987; squirrel monkey: Adams and Horton, 2003a). This results in a higher degree of anisotropy in the representation of the periphery (Fig. 5D). It could be hypothesized that the monocular crescent (the region of V1 that represents the far periphery of the visual field, receiving information only from one eye) requires fewer neurons for visual processing of a point in the visual field—perhaps as few as half of those allocated to a point well within the binocular overlap region. Moreover, it is known that the transition between the binocular and monocular regions of V1 is not abrupt. Rather, there is a gradual change from nearly equal representation of the two eyes to heavy dominance of the contralateral eye (Rosa et al., 1992). This gradual change, which is evident from eccentricities beyond 20–30°, is likely to be one of the factors that determine the shape of the areal magnification factor function of V1 in the peripheral representation (Fig. 1; see also Gattass et al., 1987). The present results indicate that the deviation of M_A from a power function (which would be evidenced by a straight line, in double-logarithmic plot such as those shown in Fig, 4A) is typically achieved through a more marked reduction in the M_E values, in comparison with M_P values, and a concomitant increase in anisotropy in the peripheral representation.

Relationship to retinal ganglion cell density

When compared with the best available estimates of ganglion cell density in the marmoset (Wilder et al., 1996), our measurements of the areal cortical magnification factor in V1 reveal a different distribution, with much greater emphasis on central vision (Fig. 7C; Table 2). It is usually the case that estimates of magnification factor and ganglion cell density are the least precise in the fovea, due to the small receptive fields and the displacement of ganglion cells to the foveal rim (e.g., Schein and de Monasterio, 1987). Nonetheless, the discrepancy between V1 and the retina is so marked that these potential sources of measurement error are unlikely to confuse the main conclusion. For example, the representation of eccentricities under 2° corresponds to 21.4% of the surface area of V1, but only 7.1% of retinal ganglion cells (Table 2). This conclusion is in agreement with results in the squirrel monkey, a species in which the central 4° of eccentricity command 27% of V1, but 12% of the ganglion cells (Adams and Horton, 2003a).

An interesting consideration in this comparison is that the marmoset fovea encompasses a relatively large proportion of the retina, compared to that of larger primate species. According to Franco et al. (2000) the fovea is similar in absolute size across species, averaging less than 0.5 mm in diameter. This results in a relatively large fovea in species with smaller eyes; the area of the marmoset retina is ~5 times smaller than that of the human eye, which results in a fovea that is proportionally 2.4 times larger. In addition, as demonstrated by Wilder et al. (1996), the ganglion cell density in the marmoset is similar to that in the much larger macaque near the fovea, but decreases more steeply with eccentricity. Nonetheless, the visuotopic map of V1 proved to be remarkably consistent across various simian species in terms of relative emphasis on central vision, and the pattern of dependence of M_A on eccentricity, although variations in other parameters (e.g., degree of anisotropy, and consequently elongation of the map) are likely to exist. Finally, it was not clear from our measurements whether the dependence of magnification factor on eccentricity in the LGN is similar to that we measured in V1. Estimates of M in the LGN are complex, due to the inherently 3D nature of this structure, with layers that vary in thickness depending on eccentricity, and may interdigitate (Spatz, 1978; Connolly and Van Essen, 1984). At the moment, the data available for the marmoset lack the resolution needed to answer this question.

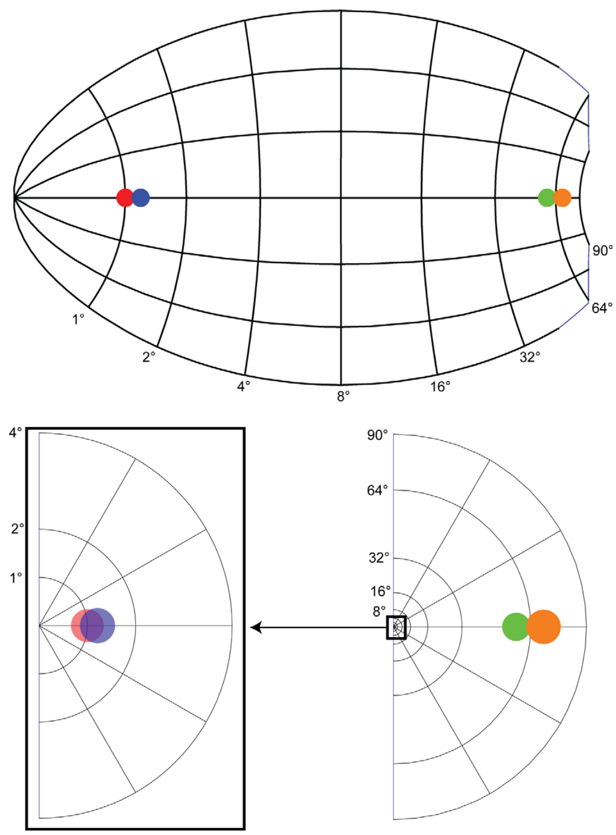

Figure 9. Diagram showing the implications of varying point image size. Top: The position of two adjacent foveal cells (red and blue) and two peripheral cells (green and orange) are shown on an idealized V1 flat map, generated by the method described by Schira et al. (2010). Bottom: The expected position and size of the receptive fields. For a similar cortical distance, the foveal receptive fields overlap considerably more than those in the periphery. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

Point image size

The present results confirm previous assessments that the point image size in V1 is highest near the fovea, and decreases towards the periphery (Dow et al., 1981; Gattass et al., 1987). Even outside the fovea, where the imprecision derived from delimiting very small receptive fields is a likely source of experimental error, there is a significant decrease in the cortical receptive field images in the cortex (Capuano and McIlwain, 1981), which form the basis of our estimates of point image size (Fig. 8A). The results we obtained in two animals are very consistent, in that they indicate that the linear distance required for mapping nonoverlapping receptive fields, measured parallel to the surface of V1, is three times larger at the fovea than at an eccentricity of 30°. Conversely, if one makes the assumption that the V1 map is such that receptive field position changes gradually and smoothly as a function of cortical location (McLoughlin and Schiessl, 2006), one would expect that the receptive fields of neurons separated by a

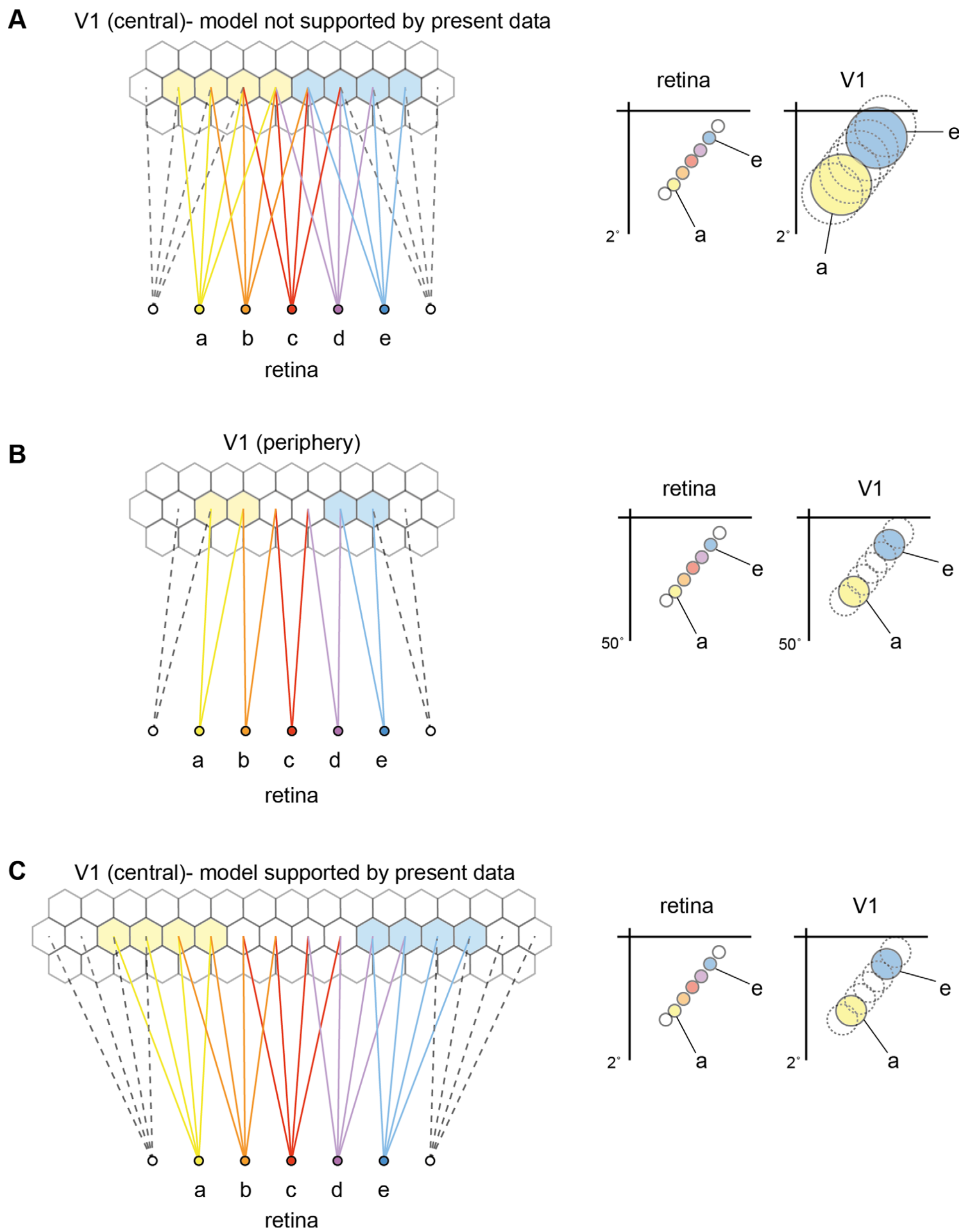

Figure 10. Different models of the anatomical relationship between retinal cells and cortical columns (hexagons). Each diagram illustrates retinal cells representing adjacent but nonoverlapping points in space (a–e, bottom) and their projections (via LGN) to V1. In theory, the increased point image size in central V1 could be a result simply of increased convergence between retinal and cortical cells, with each cortical neuron receiving information from larger number of retinal cells near the fovea (**A**) than near the periphery (**B**). This would result in relatively larger cortical receptive fields, and consequently larger overlap (shown on the right, for the representations of points a and e). Instead, our results indicate that each cortical cell receives from a nearly constant number of retinal cells (diagrams **B**,**C**). In this model the increase in point-image size in central V1 is a consequence of greater divergence in along the retinogeniculocortical projection. Both models carry the prediction that the anatomical spread of activity elicited from single retinal cells will be larger in the representation of the central vision. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

similar distance overlap more extensively centrally, than peripherally (Fig. 9). Recent studies have described approximately constant point image sizes in V1 using MRI in humans (Harvey and Dumoulin, 2011) and voltage-sensitive dye in macaques (Chen et al., 2012). However, these studies have been limited to a limited range of eccentricities (1–5° eccentricity), which may not be sufficient to reveal a significant centroperipheral trend; indeed, a slight trend towards decreasing point image size is apparent in the estimates of Harvey and Dumoulin (2011), even in this restricted range.

Our estimates of point image size (Fig. 8A) need to be seen as minimum estimates, corresponding to the diameter of a cortical region in which the vast majority of cells include a certain point in visual space. In reality, point image sizes are more precisely conceptualized in terms of statistical distributions, when one considers that receptive fields in different layers have different areas, and that there is measurable scatter even along a cellular column (Dow et al., 1981; Rosa and Tweedale, 2004). However, studies in both striate (Hubel and Wiesel, 1974) and extrastriate (Albright and Desimone, 1987) areas have shown no evidence of a variable relationship between receptive field size and scatter, across different eccentricities. The actual cortical regions activated by point-like visual stimuli are likely to be larger than indicated in Figure 8A, particularly in the supragranular and infragranular layers, and to lack sharp boundaries. Moreover, the exact extents of the receptive fields of V1 cells are known to vary according to the context in which the visual stimulus is presented (Fiorani et al., 1992; Pettet, 1992), implying that the exact extent of the point image size is malleable.

The decrease in point image size with eccentricity in V1 could result from different types of anatomical relationships along the retino–geniculo–striate pathway (Fig. 10). One mechanism that seems to be ruled out by our results is a marked change in the ratio of convergence between retinal and cortical cells, which could lead to a proportionally smaller centroperipheral increase in receptive field size among V1 cells, in comparison with retinal ganglion cells (compare Fig. 10A,B). Instead, our analyses indicate that, in the marmoset, there is an approximately constant relationship between the sizes of V1 receptive fields and the density of retinal ganglion cells, in such a way that each V1 neuron samples information conveyed by a similar number of retinal neurons, from center to periphery. Considering the areal magnification factor in V1, this in turn suggests that the inputs derived from each retinal ganglion cell are distributed over a larger area of V1 near the fovea, thereby contacting more layer 4 neurons than in the periphery (Fig. 10B,C). It remains unclear whether this amplification happens in the projection from the retina to the LGN, and from the LGN to V1, or whether it is a cumulative process involving both pathways (Malpeli and Baker, 1975; Connolly and Van Essen, 1984). Systematic measurements of the spread of single geniculocortical axons across a wide range of eccentricities are not currently available for any primate species. Nonetheless, the model illustrated in Figure 9A,B is compatible with the observation that similar-sized injections of retrograde tracers in V1 label approximately similar numbers of neurons in the LGN, from the representation of the fovea to that of eccentricities beyond 30° (Livingstone and Hubel, 1988) and with results of transneuronal labeling from the retina to the cortex (Azzopardi and Cowey, 1993).

One of the open questions left by our findings is the functional significance of the expansion of the retinal projection to foveal V1. The model proposed in Figure 10B,C implies that the information derived from each retinal ganglion cell spreads to a larger number of V1 columns in the central representation than in the peripheral representation, potentially reflecting a greater heterogeneity of functional processing modules in central vision. Whereas this can be related to the need to map two eyes centrally, but not in the far periphery, this factor alone cannot explain the marked decrease in point image size within the representation of the binocular field. Rosa et al. (1991) discussed other factors that may contribute, including the functional heterogeneity of cytochrome oxidase blobs with respect to color selectivity near the fovea (Ts'o and Gilbert, 1988). In general, our results indicate that signals derived from ganglion cells representing central vision can be multiplexed to a greater degree than those derived from peripheral ganglion cells.

CONCLUSION

We have presented an accurate measurement of cortical magnification in the marmoset V1. We conclude that V1 areal magnification is highly consistent between individuals of the same species and between different primate species, despite some differences in V1 shape and geometry. Our results argue against a strict "peripheral scaling" in the retinal representation in the cortex (Azzopardi and Cowey, 1996), but introduce a new factor that is relevant for understanding the exact nature of the geometric relationships along this pathway, by showing that the receptive fields of V1 cells represent sampling from a nearly constant number of ganglion cells across the visual field.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

ROLE OF AUTHORS

All authors had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Study concept and design: H.-H.Y. and M.G.P.R.; Acquisition of data: T.C., H.-H.Y., and M.G.P.R.; Analysis and interpretation of data: T.C., H.-H.Y., and M.G.P.R.; Drafting of the article: T.C., H.-H.Y., and M.G.P.R.; Obtained funding: M.G.P.R.

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